

1. MC.10.01

Standards set by engineering studies

- ☐ a. can determine the most efficient way of operating.
- ☐ b. can provide rigorous guidelines.
- ☐ c. may not be achievable by operating personnel.
- ☐ d. often do not allow operating personnel to have much input.
- ☒ e. All of these.



Feedback

Check My Work

Engineering studies provide efficient approaches with rigorous guidelines.

Solution
e

2. MC.10.02.ALGO (Algorithmic)

A currently attainable standard is one that

- ☐ a. uses only historical experience.
- ☐ b. relies on perfect skill in production processes.
- ☐ c. relies on maximum efficiency.
- ☐ d. assume no machine breakdowns.
- ☒ e. None of these.



Feedback

Check My Work

A currently attainable standard is rigorous but achievable and reflects a reasonable level of efficiency.

Solution
e

3. MC.10.03

An ideal standard is one that

- ☐ a. uses only historical experience.
- ☒ b. relies on maximum efficiency.
- ☐ c. can be achieved under efficient operating conditions.
- ☐ d. makes allowances for normal breakdowns, interruptions, less than perfect skill, and so on.
- ☐ e. None of these.



Feedback

Check My Work

Ideal standards can be achieved only if everything operates perfectly. No machine breakdowns, slack, or lack of skill (even momentarily) are allowed.

Solution

4. MC.10.04.ALGO (Algorithmic)

The underlying details for the standard cost per unit are provided in

- ✓ ☒ a. the standard cost sheet.
- ☐ b. the standard work-in-process account.
- ☐ c. the balance sheet.
- ☐ d. the standard production budget.
- ☐ e. None of these.

Feedback

Check My Work

This document provides the standard input prices and the standard input allowed per unit of output. It also gives the standard cost per unit.

Solution

a

5. MC.10.05

The standard cost per unit is an example of the following data analytics type:

- ☐ a. preliminary.
- ☐ b. diagnostic.
- ☐ c. predictive.
- ☐ d. descriptive.
- ✓ ☒ e. prescriptive.

Feedback

Check My Work

Correct

Solution

e

6. MC.10.06.ALGO (Algorithmic)

The standard quantity of materials allowed is computed as

- ☐ a. Unit Quantity Standard $\times$ Break-even Output.
- ☐ b. Unit Quantity Standard $\times$ Standard Output.
- ✓ ☒ c. Unit Quantity Standard $\times$ Actual Output.
- ☐ d. Unit Quantity Standard $\times$ Normal Output.
- ☐ e. None of these.

Feedback

Check My Work

Think about how you would calculate the materials that should be used for the units the company has produced.

Solution

7. MC.10.07.ALGO (Algorithmic)

The standard direct labor hours allowed is computed as

- ✓
- ☐ a. Unit Labor Standard $\times$ Practical Output.
 - ☐ b. Unit Labor Standard $\times$ Standard Output.
 - ☐ c. Unit Labor Standard $\times$ Theoretical Output.
 - ☐ d. Unit Labor Standard $\times$ Break-even Output.
 - ☒ e. Unit Labor Standard $\times$ Actual Output.

Feedback

Check My Work

Think about how you would calculate the hours that should be used for the units produced—given the standards set by the organization.

Solution
e

8. MC.10.08

Investigating variances from standard is

- ✓
- ☐ a. always done.
 - ☐ b. done if the variance is inside an acceptable range.
 - ☐ c. not done if the variance is expected to recur.
 - ☒ d. done if the variance is outside the control limits.
 - ☐ e. None of these.

Feedback

Check My Work

Random deviations from variances are not investigated.

Solution
d

9. MC.10.09

Responsibility for the materials price variance typically belongs to

- ✓
- ☐ a. production.
 - ☒ b. purchasing.
 - ☐ c. marketing.
 - ☐ d. personnel.
 - ☐ e. the chief executive officer (CEO).

Feedback

Check My Work

Logically, it should be someone associated with materials and what they cost.

Solution
b

10. MC.10.10

The materials price variance is usually computed

✓

- ☐ a. when goods are finished.
- ☐ b. when materials are issued to production.
- ☒ c. when materials are purchased.
- ☐ d. after suppliers are paid.
- ☐ e. None of these.

Feedback

Check My Work

Think about the importance of timeliness of information for control purposes.

Solution
c

11. MC.10.11

Responsibility for the materials usage variance is usually assigned to

✓

- ☐ a. the chief executive officer (CEO).
- ☐ b. marketing.
- ☐ c. purchasing.
- ☐ d. personnel.
- ☒ e. production.

Feedback

Check My Work

Logically, it should be someone who is associated with using the materials.

Solution
e

12. MC.10.12

Responsibility for the labor rate variance typically is assigned to

✓

- ☒ a. production.
- ☐ b. labor markets.
- ☐ c. personnel.
- ☐ d. labor unions.
- ☐ e. engineering.

Feedback

Check My Work

Most labor rate variances are caused by using more skilled labor in less-skilled tasks. Who decides this?

Solution
a

13. MC.10.13

Responsibility for the labor efficiency variance typically is assigned to



- ☐ a. labor unions.
- ☐ b. personnel.
- ☐ c. engineering.
- ☒ d. production.
- ☐ e. outside trainers.

Feedback

Check My Work

Think about who makes the day to day decisions on direct labor usage.

Solution
d

14. MC.10.19

The total fixed overhead variance is



- ☐ the difference between budgeted and applied fixed overhead costs.
- ☒ the difference between actual and applied fixed overhead costs.
- ☐ the difference between actual and budgeted fixed overhead costs.
- ☐ the difference between budgeted fixed and variable overhead costs.
- ☐ None of these.

Feedback

Check My Work

The total fixed overhead variance uses the standard fixed overhead rate (SFOR) times the standard hours allowed for the actual output (SH).

Post-Submission

The total fixed overhead variance is the difference between actual fixed overhead and applied fixed overhead, where applied fixed overhead is obtained by multiplying the standard fixed overhead rate (SFOR) times the standard hours allowed for the actual output (SH).

Applied FOH = SH × SFOR

Total FOH Variance = Actual FOH – Applied FOH

Solution
b

15. MC.10.20

A lean accounting manufacturing firm would most likely use which of the following?

- ☐ Static price and quantity standards.



- ☐
- ☐
- ☒
- ☐

The previous year's per unit quantity and price per unit.

A ten percent reduction in last year's standard quantity per unit and price.

Kaizen costing.

None of these.

Feedback

Check My Work
Correct
Solution
d



- ☐
- ☐
- ☐
- ☒
- ☐

too much finished goods inventory was held.

the actual output was greater than expected or practical capacity.

the company overproduced.

the actual output was less than expected or practical capacity.

All of these.

Feedback

Check My Work

The volume variance is a measure of unused capacity.

Post-Submission

The volume variance indicates the difference between what a company *budgets* for fixed overhead and *actual* amount of manufacturing costs used.

Essentially, an unfavorable volume variance indicates that output was less than budgeted.

Solution
d



- ☐
- ☐
- ☐
- ☒

a. engineering studies.

b. input from operating personnel.

c. historical experience.

d. All of these are correct.

Feedback

Check My Work
Correct
Post-Submission

Three potential sources of quantitative standards are historical experience, engineering studies, and input from operating personnel.

18. MC.10.51

Which of the following is true of ideal standards?

- ✓
- ☐ a. Ideal standards can be achieved under inefficient operating conditions.
 - ☒ b. Ideal standards demand maximum efficiency and no slack is allowed.
 - ☐ c. Ideal standards provide allowance for normal breakdowns and interruptions.
 - ☐ d. All of these are correct.

Feedback

Check My Work
Correct
Post-Submission

Ideal standards demand maximum efficiency and can be achieved only if everything operates perfectly. No machine breakdowns, slack, or lack of skill (even momentarily) are allowed.

19. MC.10.53

Which of the following is true of currently attainable standards?

- ✓
- ☐ a. They can be achieved only if everything operates perfectly.
 - ☒ b. They provide allowance for normal breakdowns, interruptions, etc.
 - ☐ c. They do not provide any allowance for any lack of skill.
 - ☐ d. They demand the maximum efficiency of all the producing factors.

Feedback

Check My Work
Correct
Post-Submission

Currently attainable standards can be achieved under efficient operating conditions. Allowance is made for normal breakdowns, interruptions, less than perfect skill, and so on. These standards are demanding but achievable.

20. MC.10.54

Standard cost systems are adopted:

- ✓
- ☐ a. to improve planning and control.
 - ☐ b. to facilitate product costing.
 - ☐ c. to enhance the operational control of firms that emphasize continuous improvement.
 - ☒ d. to improve planning and control and to facilitate product costing.

Feedback

Check My Work
Correct
Post-Submission

The benefits of operational control in a standard cost system may not extend to the manufacturing environments that emphasize continuous improvement and just-in-time purchasing and manufacturing.

21. MC.10.55

Standard cost systems can enhance operational control through the use of:

- ✓
- ☐ a. standard costs, which indicate the desired cost of a unit of input.
 - ☒ b. efficiency variances, which indicate the need for corrective action.
 - ☐ c. actual costs, which indicate the price received for units sold.
 - ☐ d. price variances, which indicate the need for better spending control.

Feedback

Check My Work
Correct
Post-Submission

Efficiency variances provide specific signals regarding the need for corrective action and where that action should be focused. Thus, in principle, the use of efficiency variances enhances operational control.

22. MC.10.68

A company manufactures model airplanes. During the month, it manufactured 25,000 airplanes. Each one used an average of 8.0 direct labor hours and an average of 2 sheets of aluminum. It normally manufactures 8,000 airplanes. Materials and labor standards for making the airplanes are:

Direct materials (1 sheet of aluminum @ \$12)	\$12
Direct materials (other accessories @ \$9)	9
Direct labor (5 hours @ \$5)	25.00

Compute the standard number of sheets of aluminum allowed for a volume of 25,000 airplanes.

- ✓
- ☐ a. 7,500 sheets
 - ☒ b. 25,000 sheets
 - ☐ c. 10,000 sheets
 - ☐ d. 20,000 sheets

Feedback

Check My Work
Correct
Post-Submission

$SQ = \text{Unit Quantity Standard} \times \text{Actual Output}$
 $SQ = 25,000 \times 1 = 25,000 \text{ sheets}$

23. MC.10.67

A company's main material is fiber board sold by the square foot. The current cost of fiber board is \$5.00 per square foot. Delivery costs are \$0.30 per board foot, wages per hour are \$25.00, payroll costs are \$3.60 per hour, and benefits are \$5.00 per hour. How much is the direct materials cost standard per square foot?

- ✓
- ☐ a. \$4.00 per sq. ft.
- ☐ b. \$5.00 per sq. ft.
- ☒ c. \$5.30 per sq. ft.
- ☐ d. \$4.75 per sq. ft.

Feedback

Check My Work
Correct
Post-Submission

Direct Materials Cost Standard per Square Foot = \$5.00 + \$0.30 = \$5.30

24. MC.10.69

Which of the following is the difference between the actual cost of materials and the materials cost allowed for the actual level of activity?

- ✓
- ☐ a. Total materials cost
- ☒ b. Total materials variance
- ☐ c. Total materials margin
- ☐ d. Total materials regression

Feedback

Check My Work
Correct
Post-Submission

The total materials variance can be due to a difference between actual and planned prices or between actual and standard quantities or both and is defined as the difference between the actual cost of materials and the materials cost allowed for the actual level of activity.

25. MC.10.73

Which of the following formulas computes usage (efficiency) variance?

- ✓
- ☐ a. (Actual Quantity – Fixed Quantity) × Standard Price Per Unit
- ☐ b. (Actual Quantity – Average Quantity) × Standard Price Per Unit
- ☐ c. (Actual Quantity – Variable Quantity) × Standard Price Per Unit
- ☒ d. (Actual Quantity – Standard Quantity) × Standard Price Per Unit

Feedback

Check My Work
Correct
Post-Submission

The usage (quantity) variance is called an efficiency variance. Usage (efficiency) variance is the difference between the actual and standard quantity of inputs multiplied by the standard unit price of the input.

26. MC.10.79

A company's standard cost is \$275,000. The allowable deviation is ±10%. Its actual costs for six months are as follows:

January	\$235,000
February	220,000
March	245,000
April	265,000
May	270,000
June	280,000

The upper and lower control limits are _____, respectively.

-  a. \$250,000 and \$225,000
-  b. \$275,000 and \$250,000
-  c. \$302,500 and \$247,500
-  d. \$305,000 and \$195,000



Feedback

Check My Work
Correct
Post-Submission

The upper control limit is \$302,500 [$\$275,000 + (\$275,000 \times 10\%)$].

The lower control limit is \$247,500 [$\$275,000 - (\$275,000 \times 10\%)$].

27. MC.10.78

A company's standard cost is \$700,000. The allowable deviation is ±5%. Its actual costs for three months are as follows:

January	\$600,000
February	560,000
March	620,000

The upper and lower control limits are _____, respectively.

-  a. \$787,500 and \$712,500
-  b. \$735,000 and \$665,000
-  c. \$742,000 and \$670,000
-  d. \$684,200 and \$557,000



Feedback

Check My Work
Correct
Post-Submission

Upper Control Limit = Standard Cost + (Standard Cost × Allowable Deviation) = [$\$700,000 + (\$700,000 \times 5\%)$] = \$735,000

Lower Control Limit = Standard Cost – (Standard Cost × Allowable Deviation) = [\$700,000 – (\$700,000 × 5%)] = \$665,000

28. MC.10.80

A company's standard cost is \$510,000. The allowable deviation is ±15%. Its actual costs for six months are as follows:

January	\$450,000
February	330,000
March	412,000
April	467,000
May	543,000
June	590,000

The actual cost that is higher than the upper control limit is:

☐

a. \$330,000.

☒

b. \$590,000.

☐

c. \$543,000.

☐

d. \$450,000.

Feedback

Check My Work

Correct

Post-Submission

Upper Control Limit = Standard Cost + (Standard Cost × Allowable Deviation)

= \$510,000 + (\$510,000 × 15%) = \$586,500 = Upper Control Limit

The actual cost that is higher than the upper control limit is \$590,000.

29. MC.10.83

Which of the following formulas is used to calculate the materials price variance?

☒

a. (Actual Price × Actual Quantity) – (Standard Price × Actual Quantity)

☐

b. (Variable Price × Actual Quantity) – (Total Price × Actual Quantity)

☐

c. (Actual Price × Actual Quantity) – (Standard Price × Standard Quantity)

☐

d. (Fixed Price × Average Quantity) – (Standard Price × Actual Quantity)

Feedback

Check My Work

Correct

Post-Submission

MPV = (AP – SP) × AQ

30. MC.10.84

Which of the following mathematical expressions is used to compute the materials usage variance?

☐

a. (Average Price × Average Quantity) - (Standard Price × Standard Quantity)

☒

b. (Standard Price × Actual Quantity) - (Standard Quantity × Standard Price)

☐

c. (Actual Price × Standard Quantity) + (Standard Price × Actual Quantity)

☐

d. (Average Price × Standard Quantity) + (Variable Price × Actual Quantity)

Feedback

Check My Work

Correct

Post-Submission

$$\text{MUV} = (\text{AQ} - \text{SQ}) \times \text{SP}$$

31. MC.10.92

During August, a company produced 15,000 chainsaw blades. The standard quantity of materials allowed was 2 pounds of steel per blade at a standard cost of \$7 per pound. The actual cost was \$7 per pound. The actual pounds of steel that the company purchased were 29,500 pounds. All materials purchased were used. Calculate the company's materials usage variance.

☒

a. \$3,500 F

☐

b. \$3,500 U

☐

c. \$4,000 U

☐

d. \$4,000 F

Feedback

Check My Work

Correct

Post-Submission

$$15,000 \text{ blades} \times 2 \text{ pounds} = 30,000 \text{ pounds of steel}$$

$$\text{MUV} = (\text{AQ} - \text{SQ}) \times \text{SP}$$

$$\text{MUV} = (29,500 - 30,000) \times \$7$$

$$\text{MUV} = \$3,500 \text{ F}$$

32. MC.10.91

During June, a company produced 12,500 chainsaw blades. The standard quantity of materials allowed was 1.5 pounds of steel per blade at a standard cost of \$8 per pound. The actual cost was \$7 per pound. The actual pounds of steel that the company purchased were 19,500 pounds. All materials purchased were used. Calculate the company's materials usage variance.

☐

a. \$6,000 F

☒

b. \$6,000 U

☐

c. \$12,000 F

☐

d. \$12,000 U

Feedback

Check My Work

Correct

Post-Submission

12,500 blades × 1.5 pounds = 18,750 pounds of steel

$$\text{MUV} = (\text{AQ} - \text{SQ}) \times \text{SP}$$

$$\text{MUV} = (19,500 - 19,750) \times \$8$$

$$\text{MUV} = \$6,000 \text{ U}$$

33. MC.10.95

A company uses a standard costing system. The following information pertains to direct materials for July:

Standard price per lb.	\$22
Actual purchase price per lb.	\$18
Quantity purchased	2,000 lbs.
Quantity used	1,400 lbs.
Standard quantity allowed for actual output	1,450 lbs.
Actual output	500 units

The company reports its materials price variances at the time of purchase. What is the materials usage variance for the company?

- ☐ a. \$1,100 U
- ☐ b. \$2,450 U
- ☐ c. \$2,450 F
- ☒ d. \$1,110 F



Feedback

Check My Work

Correct

Post-Submission

$$\text{Materials Usage Variance} = (\text{Actual Quantity} - \text{Standard Quantity}) \times \text{Standard Price} = (1,400 - 1,450) \times \$22 = \$1,100 \text{ F}$$

34. MC.10.97

During September, 50,000 units were produced. The standard quantity of materials allowed per unit was 5 pounds at a standard cost of \$2.50 per pound. If there was a favorable usage variance of \$25,000 for September, the actual quantity of materials used must have been:

- ☐ a. 105,000 pounds.
- ☐ b. 210,000 pounds.
- ☐ c. 190,000 pounds.
- ☒ d. 240,000 pounds.



Feedback

Check My Work

Correct

Post-Submission

$$50,000 \times 5 \times \$2.50 = \$625,000$$

$$\$625,000 - \$25,000 = \$600,000$$

$$\$600,000 \div \$2.50 = 240,000 \text{ pounds}$$

35. MC.10.105

A company produces picture frames. During the year, 190,000 picture frames were produced. Materials and labor standards for producing the picture frames are as follows:

Direct materials (2 pieces of wood @ \$2.25)	\$4.50
Direct labor (2 hours @ \$10)	20.00

The company purchased and used 300,000 pieces of wood at \$2 each, and its actual labor hours were 350,000 hours at a wage rate of \$11.

What is the materials usage variance?



- ☐ a. \$112,500 U
- ☒ b. \$160,000 F
- ☐ c. \$112,500 F
- ☐ d. \$160,000 U

Feedback

Check My Work
Correct
Post-Submission

Materials Usage Variance = (Actual Quantity – Standard Quantity) × Standard Price
= [300,000 – (190,000 × 2 pieces)] × \$2 = (300,000 – 380,000) × \$2 = \$160,000 F

36. MC.10.104

A company produces picture frames. During the year, 190,000 picture frames were produced. Materials and labor standards for producing the picture frames are as follows:

Direct materials (2 pieces of wood @ \$2.25)	\$4.50
Direct labor (2 hours @ \$10)	20.00

The company purchased and used 400,000 pieces of wood at \$1.90 each, and its actual labor hours were 360,000 hours at a wage rate of \$10.50.

What is the materials price variance?



- ☐ a. \$100,000 F
- ☐ b. \$112,500 U
- ☒ c. \$140,000 F
- ☐ d. \$170,000 U

Feedback

Check My Work
Correct
Post-Submission

$$\begin{aligned} \text{MPV} &= (\text{AP} - \text{SP}) \times \text{AQ} \\ (1.90 - \$2.25) \times 400,000 \\ \$140,000 \text{ F} \end{aligned}$$

37. MC.10.111

Which of the following is not true of the use of labor variance information?

- ✓ ☒ a. The actual wage rate is almost always different from the standard rate.
- ☐ b. Unexpected overtime can cause variation in the labor rate.
- ☐ c. An average wage rate is chosen as the labor rate standard.
- ☐ d. The production manager has no control over the use of labor.

Feedback

Check My Work
Correct
Post-Submission

Labor rates are largely determined by such external forces as labor markets and union contracts. The actual wage rate rarely departs from the standard rate.

38. MC.10.114

A company uses a standard costing system. The following information pertains to direct labor costs for February:

Standard direct labor rate per hour	\$15.00
Actual direct labor rate per hour	\$13.50
Labor rate variance	\$19,500 F
Actual output	1,000 units
Standard hours allowed for actual production	10,000 hours

What is the total labor variance for the company?

- ✓ ☒ a. \$12,000 F
- ☐ b. \$12,000 U
- ☐ c. \$25,500 F
- ☐ d. \$25,500 U

Feedback

Check My Work
Correct
Post-Submission

$$\begin{aligned} \$19,500 \div (\$15.00 - \$13.50) &= 13,000 \text{ actual hours} \\ (13,000 \times \$13.50) - (10,000 \times \$15) &= \$25,500 \text{ U} \end{aligned}$$

39. MC.10.122

A corporation produced 100 units of Product AA. The total standard and actual costs for materials and direct labor for the 100 units of Product AA are as follows:

Materials:	Standard	Actual
Standard: 300 pounds at \$2.50 per pound	\$750	
Actual: 250 pounds at \$3.00 per pound		\$750
Direct labor:		
Standard: 400 hours at \$10.00 per hour	4,000	
Actual: 350 hours at \$12.00 per hour	4,200	

What is the labor efficiency variance for the corporation?

- ☐ a. \$602 U
- ☐ b. \$400 F
- ☐ c. \$425 U
- ☒ d. \$500 F

Feedback

Check My Work
Correct
Post-Submission

$$\text{Labor Efficiency Variance} = (\text{Actual Hours} - \text{Standard Hours}) \times \text{Standard Rate} = (350 - 400) \times \$10 = \$500 \text{ F}$$

40. MC.10.125

A company constructs houses. The standard labor rate is \$25 per hour, and the standard number of hours is 15,000 hours per house. During the year, it constructed 10 houses using 18,000 labor hours per house at a rate of \$28 per hour.

Calculate the company's labor rate variance.

- ☐ a. \$648,000 U
- ☐ b. \$540,000 F
- ☒ c. \$540,000 U
- ☐ d. \$648,000 F

Feedback

Check My Work
Correct
Post-Submission

$$\begin{aligned} \text{LRV} &= (\text{AR} - \text{SR}) \times \text{AH} \\ (\$28 - \$25) \times 180,000 \\ \$540,000 \text{ U} \end{aligned}$$

41. MC.10.129

During September, a small roofing company purchased 200 bundles of a certain type of shingle at a price of \$25 per bundle, \$5 less than the standard price. Its standard quantity of this type of shingle is 400 bundles. What is the journal entry to record the purchase?

purchase of materials?



- ☐ a. Debit Materials, 18,000
Credit Materials Price Variance, 2,000
Credit Accounts Payable, 15,000
- ☐ b. Debit Materials, 2,500
Debit Materials Price Variance, 2,600
Credit Accounts Payable, 1,720
- ☒ c. Debit Materials, 6,000
Credit Materials Price Variance, 1,000
Credit Accounts Payable, 5,000
- ☐ d. Debit Materials, 5,000
Debit Materials Price Variance, 3,500
Credit Accounts Payable, 21,500

Feedback

Check My Work
Correct
Post-Submission

$$\text{Standard Price} = \text{Actual Price} + \$5 = \$25 + \$5 = \$30$$
$$\text{Materials} = \text{Standard Price} \times \text{Actual Quantity} = \$30 \times 200 = \$6,000$$
$$\text{Materials Price Variance} = (\text{Actual Price} - \text{Standard Price}) \times \text{Actual Quantity}$$
$$= (\$25 - \$30) \times 200 = \$1,000 \text{ F}$$

42. MC.10.132

During April, a small roofing company purchased 400 bundles of a certain type of shingle at a price of \$40 per bundle, \$5 more than the standard price. The standard quantity of this type of shingle is 440 bundles.

What is the journal entry to record the purchase of materials?



- ☐ a. Debit Materials, 24,500
Credit Materials Price Variance, 5,600
Credit Accounts Payable, 18,900
- ☒ b. Debit Materials, 14,000
Debit Materials Price Variance, 2,000
Credit Accounts Payable, 16,000
- ☐ c. Debit Materials, 19,575
Debit Materials Price Variance, 5,800
Credit Accounts Payable, 25,375
- ☐ d. Debit Materials, 20,000
Credit Materials Price Variance, 5,200
Credit Accounts Payable, 13,200

Feedback

Check My Work
Correct
Post-Submission

$$\text{Accounts Payable} = \text{Actual Price} \times \text{Actual Quantity} = \$40 \times 400 = \$16,000$$
$$\text{Materials Price Variance} = (\text{Actual Price} - \text{Standard Price}^*) \times \text{Actual Quantity} = (\$40 - \$35) \times 400 = \$5 \times 400 = \$2,000 \text{ U}$$
$$\text{Materials} = \text{Actual Quantity} \times \text{Standard Price}^* = 400 \times \$35 = \$14,000$$

*Standard Price = \$40 – \$5 = \$35

43. MC.10.136

A company manufactures t-shirts. During the year, it manufactured 250,000 t-shirts, using 2 hours of direct labor at a rate of \$8.50 per hour. The materials and labor standards for manufacturing the t-shirts are as follows:

Direct materials (6 yards of fabric @ \$4.00 per yard)	\$24.00
Direct labor (2.4 hours @ \$8.00 per hour)	19.20

It took the company 1,400,000 yards at \$2.75 per yard to make the 250,000 t-shirts.

What is the company's materials price variance, assuming that materials purchased equals materials used?

- ✓
- ☒

a. \$350,000 F
- ☐

b. \$350,000 U
- ☐

c. \$700,000 U
- ☐

d. \$700,000 F

Feedback

Check My Work
Correct
Post-Submission

$$\begin{aligned} \text{MPV} &= (\text{AP} - \text{SP}) \times \text{AQ} \\ &= (\$2.75 - \$3.00) \times 1,400,000 \\ &= \$350,000 \text{ F} \end{aligned}$$

44. MC.10.144

In a standard cost system, variable overhead is applied using:

- ✓
- ☐

a. budgeted indirect labor hours.
- ☐

b. direct labor hours at practical capacity.
- ☒

c. standard direct labor hours.
- ☐

d. actual direct labor hours.

Feedback

Check My Work
Correct
Post-Submission

Variable overhead is applied using hours allowed in a standard cost system.

45. MC.10.147

The formula for calculating the variable overhead efficiency variance is:

- ✓
- ☐

a. $(-AH - SH) \times AVOR$.
- ☐

b. $(-AVOR \times AHS) \times SVOR$.
- ☒

c. $(AH - SH) \times SVOR$.
- ☐

d. $(AVOR - SHOR) \times SH$.

Check My Work
Correct
Post-Submission

$$\text{VOH Efficiency Variance} = (\text{AH} - \text{SH}) \times \text{SVOR}$$

46. MC.10.150

Responsibility for the variable overhead spending variance is usually assigned to the:

- ✓
- ☐ a. engineering department.
 - ☒ b. production department.
 - ☐ c. personnel department.
 - ☐ d. purchasing department.

Check My Work
Correct
Post-Submission

Responsibility for the variable overhead spending variance is generally assigned to production departments.

47. MC.10.158

Fixed overhead was budgeted at \$200,000, and 25,000 direct labor hours were budgeted. If the fixed overhead volume variance was \$8,000 favorable and the fixed overhead spending variance was \$6,000 unfavorable, fixed overhead applied must be:

- ✓
- ☐ a. \$202,000.
 - ☐ b. \$194,000.
 - ☐ c. \$206,000.
 - ☒ d. \$208,000.

Check My Work

SUPPORTING CALCULATIONS:

$$\$200,000 + \$8,000 = \$208,000$$

48. MC.10.162

Thermal Company's standard fixed overhead rate is \$6.00 per direct labor hour based on budgeted fixed costs of \$600,000. The standard allows 1 direct labor hour per unit. During the year, Thermal produced 110,000 units of product, incurred \$630,000 of fixed overhead costs, and recorded 212,000 actual hours of direct labor.

What is the activity level on which Thermal based its fixed overhead rate?

- ☐ a. 50,000 direct labor hours
- ☐ b. 110,000 direct labor hours
- ☐ c. 105,000 direct labor hours

✓ ☐ d. 100,000 direct labor hours

Feedback

Check My Work
Correct
Post-Submission

SUPPORTING CALCULATIONS:

$$\$600,000 \div \$6.00 = 100,000 \text{ DLH}$$

49. MC.10.165

If variable manufacturing overhead is applied based on direct labor hours and there is an unfavorable direct labor efficiency variance the:

- ✓ ☐ a. direct labor rate variance will be favorable.
- ☒ b. variable manufacturing overhead efficiency variance will be unfavorable.
- ☐ c. variable manufacturing overhead spending variance will be unfavorable.
- ☐ d. direct materials usage variance will be unfavorable.

Feedback

Check My Work
Correct
Post-Submission

If variable manufacturing overhead is applied based on direct labor hours and there is an unfavorable direct labor efficiency variance, the variable manufacturing overhead efficiency variance will be unfavorable.

50. MC.10.167

Gluten Corporation uses a standard costing system. Information for the month of May is as follows:

Actual manufacturing overhead costs (\$24,000 is fixed) \$80,000

Direct labor:

Actual hours worked	12,000 hrs.
Standard hours allowed for actual production	10,000 hrs.
Average actual labor cost per hour	\$18.00

The factory overhead rate is based on a normal volume of 11,000 direct labor hours. Standard cost data at 12,000 direct labor hours were as follows:

Variable overhead	\$48,000
Fixed factory overhead	24,000
Total factory overhead	<u>\$72,000</u>

What is the variable overhead efficiency variance for Gluten?

- ☐ a. \$4,000 U
- ☐ b. \$20,000 U



c. \$2,000 U



d. \$8,000 U

Feedback

Check My Work

Correct

Post-Submission

SUPPORTING CALCULATIONS:

$$(12,000 - 10,000) \times (\$48,000 \div 12,000) = \$8,000 \text{ U}$$

51. MC.10.171

King Company calculates its predetermined rates using practical volume, which is 325,000 units. The standard cost system allows 3 direct labor hours per unit produced. Overhead is applied using direct labor hours. The total budgeted overhead is \$4,260,000, of which \$994,000 is fixed overhead. The actual results for the year are as follows:

Units produced: 318,000

Direct labor: 965,000 hours @ \$12.00/hour

Variable overhead: \$3,302,000

Fixed overhead: \$998,000

(Round overhead rates to two decimal places.)

Calculate the variable overhead spending variance.



a. \$69,250 F



b. \$40,000 F



c. \$24,000 U



d. \$69,250 U

Feedback

Check My Work

Correct

Post-Submission

$$\text{Spending Variance} = \text{AVOH} - (\text{SVOR} \times \text{AH}) = \$3,302,000 - (\$3.35^* \times 965,000) = \$69,250 \text{ U}$$

$$*\$4,260,000 - \$994,000 = \$3,266,000;$$

$$\$3,266,000 \div (3 \times 325,000) = \$3.35$$